

Document 9. Market Structures

Topic overview. Market structures range from Perfect Competition (PC) to pure Monopoly. The structure determines firm behaviour — output, pricing, non-price strategies — and the resulting allocative, productive and dynamic efficiency.

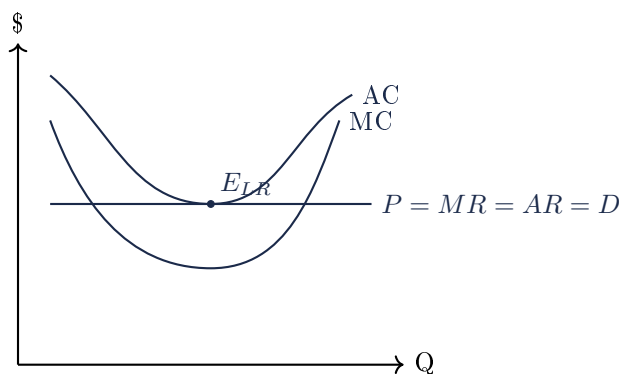
Section A. Important Definitions

TR, TC, AR, MR, AC, MC	Total Revenue, Total Cost, Average / Marginal versions.
Profit-maximisation condition	Produce where $MR = MC$ with MC cutting MR from below.
Normal profit	The minimum return required to keep resources in the industry: $AR = AC$.
Supernormal profit	$AR > AC$ at the profit-max quantity.
Allocative efficiency	$P = MC$: the resources are allocated to the goods most valued by society.
Productive efficiency	Produced at minimum AC.
Dynamic efficiency	Innovation, product development, cost reduction over time.
Barriers to entry	Structural (cost, network, brand) or strategic (limit pricing, predatory pricing).

Section B. Structure Comparison Table

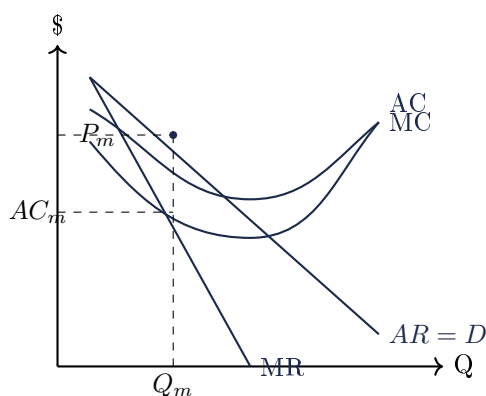
Feature	PC	Monop. Comp.	Oligopoly	Monopoly
No. of firms	Very many	Many	Few	One
Product	Homogeneous	Differentiated	Homogeneous or differentiated	Unique
Barriers to entry	None	Low	High	Very high
Knowledge	Perfect	Imperfect	Imperfect	N/A
Price-setter?	Price-taker	Some power	Considerable	Significant
Long-run profit	Normal	Normal	Supernormal possible	Supernormal
Efficiency	Allocative + productive	Neither	Neither	Neither (but maybe dynamic)

Section C. Diagram — Perfect Competition (Long Run)



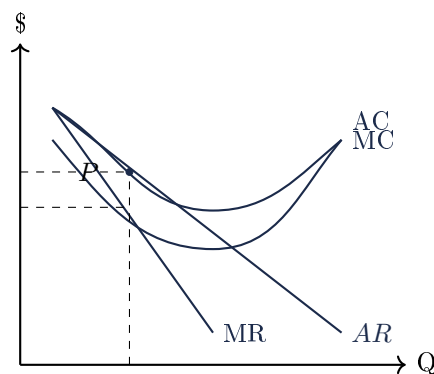
NOTE: In long-run PC equilibrium, $P = MR = AR = MC = \min(AC)$: simultaneously allocatively and productively efficient. Normal profit only.

Section D. Diagram — Monopoly

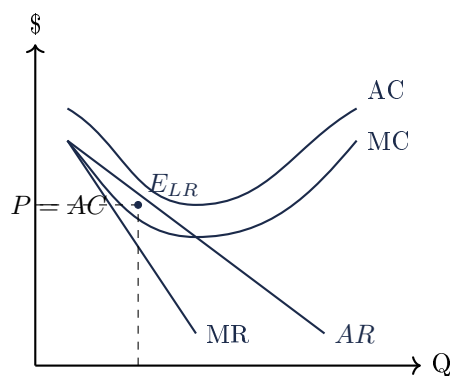


Caption. Monopolist sets $MR = MC$, then reads price off the AR curve. Supernormal profit = $(P_m - AC_m) \times Q_m$.

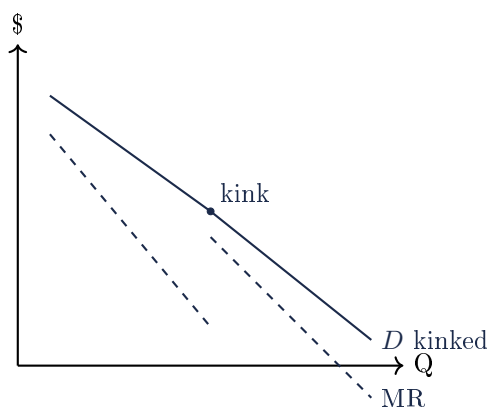
Section D1. Diagram — Monopolistic Competition (Short Run with supernormal profit)



Section D2. Diagram — Monopolistic Competition (Long Run, normal profit)



Section E. Diagram — Kinked Demand (Oligopoly)



Section F. Question Type 1: Compare firm behaviour across structures

Step 1. State the profit-max rule applies to all structures: $MR = MC$.

Step 2. For PC, $P = MR = AR$ so $P = MC$ ($\therefore$ allocatively efficient).

Step 3. For Monopoly, $MR < AR = P$ so $P > MC$ ($\therefore$ allocatively inefficient).

Step 4. For Oligopoly, kinked DD explains price rigidity; price = above MC.

Step 5. Conclude on welfare (PC best for static efficiency; Monopoly may dominate on dynamic efficiency).

Section G. Question Type 2: Explain oligopoly strategies

- **Price strategies:** kinked-demand inertia; price leadership; collusion (explicit cartel, tacit understanding); limit pricing; predatory pricing; price discrimination.
- **Non-price strategies:** advertising; branding; product differentiation; R&D; loyalty programmes; vertical integration.
- **Game theory:** Prisoners' Dilemma to show incentive to collude but instability without enforcement.

Section H. Question Type 3: Evaluate market dominance

Pro market dominance	Economies of scale lower AC; dynamic efficiency from R&D supernormal profits; international competitiveness; cross-subsidisation.
Anti market dominance	Allocative & productive inefficiency; X-inefficiency; consumer choice limited; rent-seeking and capture.

Section I. Singapore Applications

- **Monopoly / near-monopoly:** SBS Transit + SMRT public transport; SP Group electricity; SingTel/StarHub/M1 telecom oligopoly.
- **Competition regulator:** Competition and Consumer Commission of Singapore (CCCS) adjudicates anti-competitive practices.
- **Recent ruling:** 2022 CCCS opinion on Grab–Uber merger; conditional approval of Sheng Siong / supermarket consolidation.

Section J. Worked Examples

Worked Example 1: Why Singapore telcos compete on non-price strategies

With three near-equal players (Singtel, StarHub, M1) and a high entry barrier (spectrum cost), DD is kinked — competitors match price cuts but not price rises. Firms therefore compete via bundles, content, loyalty schemes (NPS). Recent entry of TPG/Simba illustrates how a regulatory carve-out can soften the kink, returning the market to price-led competition.

Worked Example 2: Should the government break up a monopoly?

Break-up improves allocative efficiency ($P \downarrow$, $Q \uparrow$) but may sacrifice scale economies and innovation. The trade-off depends on (i) cost structure, (ii) contestability, (iii) the regulator's quality. In natural monopolies (e.g. MRT), price-cap regulation dominates break-up.